

Kye Shibata

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Research Interests

Historical linguistics

- Focus: Phonological/phonetic reconstruction of Proto-Austronesian; subgrouping and contact history of Formosan languages; determining phonetic bases for irregularities in sound changes and sound correspondences
- Languages: Austronesian languages, especially Formosan languages

Articulatory phonetics

- Focus: Articulatory studies, particularly on liquids
- Methods: Electromagnetic articulography; ultrasound tongue imaging; aerodynamic analysis; acoustic analysis
- Languages: English, Mandarin Chinese, Japanese, Korean, Vietnamese, Formosan languages

Education

National Tsing Hua University

September 2020 – February 2025

Ph.D. in Linguistics

Dissertation title: “A reconstruction of Proto-Austronesian coronal consonants: Phonetic evidence from Formosan languages”

Committee: Hsiu-chuan Liao (advisor), Malcolm Ross, Alexander Adelaar, Elizabeth Zeitoun, Feng-fan Hsieh

National Tsing Hua University

September 2018 – August 2020

M.A. in Linguistics

Thesis title: “A reconstruction of Proto-Bunun phonology and lexicon”

Committee: Hsiu-chuan Liao (advisor), Hui-chuan J. Huang, Rik De Busser

State University of New York, College at Geneseo

August 2008 – May 2013

B.S. in Applied Physics

Minors: Mathematics, Computer Science

Skills

Languages

English (C2), Japanese (C2), Mandarin (Upper intermediate - TOCFL B2)

Technical Skills

Programming: Python, MATLAB, R, SQL

Equipment: Carstens AG501 (EMA), Articulate Instruments Micro System (UTI), Glottal Enterprises Aeroview/Dualview system (Aerodynamics) and Electrolottograph (EGG)

Research Experience

Institute of Linguistics, Academia Sinica

July 2025 – *Current*

Postdoctoral Fellow

PI: Elizabeth Zeitoun

- Project: Historical analysis of the Western Plains languages of Taiwan, primarily based on an analysis of primary data collected by Japanese scholars in the late nineteenth and early twentieth centuries.

National Tsing Hua University Phonetics Laboratory

February 2025 – June 2025

Postdoctoral Research Fellow

PI: Yueh-chin Chang

- Responsibilities: Design and implementation of new experimental methods in articulatory phonetics; Data analysis; Technical manager
- Projects: Quantifying lateral dynamics using a 3D palate model with EMA; Optical positional correction for ultrasound tongue imaging; Aerodynamics of implosives

National Tsing Hua University Phonetics Laboratory

January 2022 – January 2025

Research Assistant

PIs: Feng-fan Hsieh and Yueh-chin Chang

- Projects: Kinematics of Vietnamese implosives, Classification of Korean /l/, Lateral dynamics in Mandarin, Lateral dynamics in Formosan languages
- Results: Poster presentation at the 13th International Seminar on Speech Production

Institute of Linguistics, Academia Sinica

January 2023 – June 2023

Research Assistant

PI: Elizabeth Zeitoun

- Project: Research on Paiwan historical phonology

Graduate School of Engineering, The University of Tokyo

April 2014 – August 2015

Graduate Student, Electrical Engineering (non-degree)

- Project: Coil design for wireless power transfer to in-wheel motors
- Results: Oral presentation at the IEEJ International Workshop on Sensing, Actuation, and Motion Control.

Department of Physics and Astronomy, SUNY Geneseo

May 2009 – May 2013

Research Assistant, Engineering Lab

- Responsibilities: Circuit design and programming related to the Rapid Pneumatic Transportation of Radioactive Samples System (RaPToRS), including four years of full-time summer work.
- Results: Poster presentations at the 52nd and 53rd Annual Meetings of the APS Division of Plasma Physics.

Teaching Experience

University Courses

Institute of Linguistics, National Tsing Hua University

January 2020 – December 2023

Teaching Assistant

- Responsibilities: Lecturing; Leading in-class problem sessions; Overseeing students' class progress and projects
- Courses: (Graduate-level courses marked with *)
 - Field Methods* (Spring 2020, Spring 2022)
 - Historical Linguistics* (Fall 2020, Fall 2021, Fall 2022)
 - Phonology II* (Spring 2021)
 - Articulatory Phonetics* (Fall 2020)
 - Languages of the World (Spring 2021, Spring 2022, Spring 2023)
 - Linguistic Analysis (Fall 2023)
 - Language and History (Fall 2020, Fall 2022)
 - Grammatical Structures of the World's Languages (Fall 2021)

Department of Physics and Astronomy, SUNY Geneseo

January 2010 – December 2011

Laboratory Instructor

- Responsibilities: Lecturing; Leading in-class experiments; Writing, administering, and grading quizzes; Grading laboratory notebooks
- Courses:
 - The Physics of Sound (Fall 2010, Fall 2011)
 - The Physics of Light (Spring 2011)

English Teaching

HESS International Educational Group

October 2015 – July 2018

English Teacher

- Responsibilities: Teaching English to elementary school-aged children in Taiwan, mostly of a Mandarin speaking background.
- Student level: Second language speakers, many of whom attended English-only or bilingual kindergartens.

Global Communications

May 2015 – December 2016

English Instructor (part time)

- Responsibilities: 1-on-1 online English instruction to Japanese students, primarily adults of employment age.
- Student level: Varying levels of fluency.

Publications & Presentations

Journal Articles

- Shibata, Kye**, Hsieh, Feng-fan, Chang, Yueh-chin, & Tiede, Mark. (2026). Improved measurement of lateral parasagittal articulation integrating three-dimensional palate shape. *Journal of Phonetics* 116, 101492. DOI: <https://doi.org/10.1016/j.wocn.2026.101492>
- Shibata, Kye**. (2025). On the reconstruction of Proto-Austronesian voiced coronal obstruents. *Oceanic Linguistics* 64(2), 255-283. DOI: <https://dx.doi.org/10.1353/ol.2025.a974303>

Book Chapters

- Shibata, Kye**, Hsieh, Feng-fan, Huang, Hui-chuan Jennifer, McGuire, Paul, & Chang, Yueh-chin. (Under review). Articulatory characteristics of implosives in Bunun. Invited contribution in Bonnie Sands, Maida Percival, Anne-Maria Fehn, & Jeremy Coburn (eds.), *Non-Pulmonic Consonants*, University of Toronto Press.
- Shibata, Kye**, (Under review). Kinship terms in the Western Plains. Invited contribution in Elizabeth Zeitoun & Hong-sui Lim (eds.), *An exploration of kinship terminologies in the Austronesian languages of Taiwan*.

Conference Proceedings

- McGuire, Paul, **Shibata, Kye**, Cao, Viet Thanh, Hsieh, Feng-fan, & Chang, Yueh-chin. (2025). Supralaryngeal kinematics of implosives in Central Vietnamese: An EMA study. In *Proceedings of Interspeech 2025*, 3484–3488. DOI: <https://doi.org/10.21437/Interspeech.2025-1733>
- Shibata, Kye**, Hsieh, Feng-fan, & Chang, Yueh-chin. (2024). Allophones of Korean /l/: A classification using EMA. In *Proceedings of the 13th International Seminar on Speech Production*, 16–19. DOI: <https://doi.org/10.21437/issp.2024-5>
- Huang, Jing Molly, **Shibata, Kye**, Hsieh, Feng-fan, Chang, Yueh-chin, & Tiede, Mark. (2023). The L~N merger in Southwestern Mandarin: An articulatory study. In *Proceedings of the 20th International Congress of Phonetic Sciences 2023*, 1092–1096. [Equal credit given to first and second authors, listed in alphabetical order]

Conference presentations

- Shibata, Kye**. (2025). “In search of the Austronesian L: A phonetic reconstruction of Proto-Austronesian laterals.” Oral presentation at the *34th Annual Meeting of the South East Asian Linguistics Society*. Bali, Indonesia.
- Shibata, Kye**. (2024). “The history of language contact in “Tsouic” languages.” Oral presentation at the *16th International Conference on Austronesian Linguistics*. Manila, Philippines.
- Smith, Alexander D., **Shibata, Kye**, Yu, Jian, & Schuelke, Peter. (2024). “The linguistic evidence for Austronesian watercraft innovations.” Oral presentation at the *16th International Conference on Austronesian Linguistics*. Manila, Philippines.
- Shibata, Kye** & Pan, Samuel Yu-hsiang. (2024). “The reconstruction of Proto-Austronesian final *-ʔ, *-h, and *-∅.” Oral presentation at the *33rd Annual Meeting of the South East Asian Linguistics Society*. Taipei, Taiwan.

- Shibata, Kye.** (2023). “Irregularities in the reflexes of Proto-Austronesian *z and *d: a phonetic explanation.” Oral presentation at the *32nd Annual Meeting of the South East Asian Linguistics Society*. Chiangmai, Thailand.
- Shibata, Kye.** (2022). “The impact of language contact on the phonology of Paiwan.” Oral presentation at *Historical Relationships of East and Southeast Asian Languages*. Hsinchu, Taiwan. (Presented online due to the COVID-19 Pandemic)
- Shibata, Kye.** (2022). “Reexamining the historical relationships of the “Western Plains” languages.” Oral presentation at the *31st Annual Meeting of the South East Asian Linguistics Society*. Honolulu, USA. (Presented online due to the COVID-19 Pandemic)

Invited Talks

- Shibata, Kye.** (2025). “Practical, portable, and pain-free: Improving ultrasound image collection for endangered languages.” Invited talk at *the 2025 NTHU Linguistics Forum—Phonetics Workshop*, National Tsing Hua University.
- Shibata, Kye & Hsieh, Feng-fan.** (2024). “A new angle on lateralization.” Invited talk at *the 2024 NTHU Linguistics Forum (Series 2): Experimental Linguistics*, National Tsing Hua University.
- Shibata, Kye.** (2023). “Quantification techniques for lateral consonants using EMA.” Invited talk at *the Workshop on Liquid Production*, National Tsing Hua University.

Theses

- Shibata, Kye.** (2025). *A reconstruction of Proto-Austronesian coronal consonants: Phonetic evidence from Formosan languages*. Ph.D. dissertation. Institute of Linguistics, National Tsing Hua University.
- Shibata, Kye.** (2020). *A reconstruction of Proto-Bunun phonology and lexicon*. MA thesis. Institute of Linguistics, National Tsing Hua University.

Other Research Output (Non-Linguistics)

- Shibata, Kye, Imura, Takehiro & Fujimoto, Hiroshi.** (2015). “Basic Study on Coil Configurations for Direct Wireless Power Transfer from Road to Wireless In-Wheel Motor.” Oral presentation at the *IEEJ International Workshop on Sensing, Actuation, and Motion Control*. Nagoya, Japan.
- Shibata, Kye, Krieger, Michael, Fallica, Jacob, Henchen, Robert, Pogozielski, Edward, & Padalino, Stephen.** (2011). “Performance Characterization of RaPToRS Systems.” Poster presentation at the *53rd Annual Meeting of the APS Division of Plasma Physics*. Salt Lake City, USA.
- Henchen, Robert, **Shibata, Kye, Krieger, Michael, Pogozielski, Edward, Padalino, Stephen, Glebov, Vladimir, & Sangster, Craig.** (2010). “RaPToRS Sample Delivery System.” Poster presentation at the *52nd Annual Meeting of the APS Division of Plasma Physics*. Chicago, USA.